

Systems

Script: Chapter 2 (Systems)

Learning goals

- Use the operator view $y = S\{x\}$ and decide for a given system whether it is static or dynamic (memory), causal, linear and time-invariant – by proof and by experiment.
- Design numerical test harnesses that *reveal* a property and know their pitfalls (impulse-only tests, small amplitudes, boundary effects).
- Model engine subsystems as dynamic systems: inertia (first-order lag), wall film, transport delay, switching sensor.
- Understand the limit cycle of a two-point lambda controller and predict its period from the loop dead time.

Definitions used on this sheet

For a system $y = S\{x\}$ (continuous or discrete time):

- *memoryless (static)*: $y(t_0)$ depends only on $x(t_0)$;
- *causal*: $y(t_0)$ depends only on $x(t)$, $t \leq t_0$; equivalently, if $x_1(t) = x_2(t)$ for $t < t_0$, then $y_1(t) = y_2(t)$ for $t < t_0$;
- *linear*: $S\{ax_1 + bx_2\} = aS\{x_1\} + bS\{x_2\}$ for all inputs and constants (superposition = additivity + homogeneity);
- *time-invariant*: $y(t) = S\{x(t)\} \Rightarrow y(t - t_0) = S\{x(t - t_0)\}$ for every shift t_0 (discrete time: integer shifts n_0).

Linearity and time invariance are previewed here; Chapter 3 shows why their combination (LTI) is so powerful.

Exercise 2.1 Properties by hand**PAPER**

- a) Decide for each system whether it is memoryless, causal, linear and time-invariant (table with yes/no):

S1: $y(t) = 2x(t)$	S2: $y(t) = x^2(t)$	S3: $y[n] = x[n+1] - x[n]$
S4: $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$	S5: $y(t) = x(t - T_d)$, $T_d > 0$	S6: $y(t) = x(t) $
S7: $y(t) = \int_{-\infty}^t x(\tau) d\tau$	S8: $y[n] = nx[n]$	S9: $y[n] = x[n] + 0.5$
S10: $y[n] = x[2n]$	S11: $n_k = 1/T_k$ (Sheet 1)	S12: $U = 0.5 + 0.4 \tanh \frac{1-\lambda}{0.005}$

- b) Prove your answers formally for S6 (linearity), S8 (time invariance) and S10 (causality and time invariance). For S9, show that the system is not linear, but that the *difference* of two outputs depends linearly on the difference of the inputs (*incrementally linear*).
- c) An engineer computes the mean speed of one crank revolution as the arithmetic mean of the 58 per-tooth speeds $1/T_k$, treating the gap period like a normal tooth period. Compute the relative error at constant speed. After correcting the gap ($3/T_{\text{gap}}$), is the arithmetic mean of the per-tooth speeds equal to $60/T_{\text{rev}}$? Which property of S11 is responsible, and what is the error for a speed fluctuation of $\pm 1\%$ (sinusoidal)?

Setup: JSLinux

From this sheet on, most lab exercises run in JSLinux, a Linux system in the browser. If you have not done Bonus Exercise 0.B1, set it up now (about 10 min):

1. Open <https://bellard.org/jslinux>, start the *x86 Alpine Linux* (console) virtual machine and check the compiler with `gcc -version` (or `tcc -v`).
2. Upload the course headers `engine_signals.h`, `asciipplot.h`, `csvio.h` (code/common/) and the files of the exercise (e.g. `code/jslinux/sheet02/`) with the upload button below the terminal; check with `ls` that they arrived. Headers and program must be in the same directory.
3. Compile and run, e.g.

```
1 gcc -O2 -o blackbox blackbox.c -lm
2 ./blackbox
```

4. The VM forgets all files when the page is reloaded: keep your sources on your own computer and upload them again. Text from Wokwi (Serial Monitor) gets into the VM with `cat > data.csv`, paste, `Ctrl-D`; `export_file data.csv` downloads a file from the VM to your computer.

Exercise 2.2 Black-box system identification

JSLINUX

The header `code/jslinux/sheet02/mystery.h` contains five unknown systems

```
void Sk(const double *x, double *y, int N);  k = 1, ..., 5
```

(input $x[0..N-1]$, $x[n] = 0$ outside, output $y[0..N-1]$). One of them is LTI; the others violate exactly one or two of the properties of Exercise 2.1. **Do not read the header** – find the properties by experiments, as you would with a supplier’s ECU function delivered as object code. The program `code/jslinux/sheet02/blackbox.c` plots the responses to unit impulses at $n_0 = 30$ and $n_0 = 31$ and contains a (weak) homogeneity test with scaled impulses as an example.

- a) Superposition test: compare $S\{ax_1 + bx_2\}$ with $aS\{x_1\} + bS\{x_2\}$ for random signals and random weights, at amplitude 1 and 0.01. Why is the impulse test of the template not sufficient, and why two amplitudes?
- b) Shift test: compare $S\{x[n-k]\}$ with $y[n-k]$ for $k = 1 \dots 4$. Why are the first and last samples of the test signal set to zero?
- c) Causality test: two inputs that agree for $n < n_0$ and differ afterwards.
- d) Memory test: two inputs that differ only at n_0 . Why must the test be repeated for several n_0 ?
- e) Report a property table with the numerical evidence (largest deviation of each test). For the LTI systems determine $h[n]$ from the plots and give the difference equation. Why does an “impulse response” not characterise the other systems?

Exercise 2.3 The engine as a dynamic system

JSLINUX

Program: `code/jslinux/sheet02/engine_dyn.c`.

- a) *Crankshaft inertia*. With fuel quantity q (1 = nominal), the engine speed obeys $\tau \dot{n} = Kq - n$, $\tau = 0.4$ s, $K = 2000$ rpm. Simulate a fuel step $1 \rightarrow 1.5$ at $t = 0.5$ s with the Euler method ($T = 1$ ms) and with the exact discretisation $n[k] = an[k-1] + (1-a)Kq[k-1]$, $a = e^{-T/\tau}$ (derive it!). Determine τ from the 63 % point and compare both methods with the analytic solution at $t = 0.6$ s. Is the engine (fuel $\rightarrow$ speed) static or dynamic?

- b) *Lambda path.* Implement `lp_step()` (one step = 1 ms) – it is the model inside the Wokwi chip `lambda-probe`: $q \rightarrow$ wall film (first-order lag, 20 ms) $\rightarrow \lambda = (1 + a_{\text{air}})/q_f \rightarrow$ transport delay $T_d \rightarrow$ sensor lag (30 ms) $\rightarrow U = 0.5 + 0.4 \tanh((1 - \lambda)/0.005)$. Classify each block (static/dynamic, linear, TI). Simulate fuel steps $1.00 \rightarrow 1.10$ and $1.00 \rightarrow 1.20$ ($a_{\text{air}} = +5\%$, $T_d = 200$ ms) and measure the time until U crosses 0.45 V. Which observations prove that the chain is nonlinear? Under which operating condition is it not time-invariant on a real engine?
- c) *Closed loop – prediction for Exercise 2.4.* A two-point controller with integrator changes q with slope $-r$ while the sensor reports rich ($U > 0.45$ V) and $+r$ while lean (period 10 ms, 8-bit PWM). Derive the period and the λ amplitude of the resulting oscillation as functions of T_d , r and the lag time constants (hint: a ramp passing a first-order lag τ is delayed by τ). Implement the controller in `closed_loop()`, compare with your prediction for $T_d = 50 \dots 800$ ms, and explain what happens to the mean of q when the unmetered air changes. Extend the prediction to a controller with an additional jump $\pm P$ at every switching of the sensor.

Exercise 2.4 Closed lambda loop

WOKWI

Project `code/wokwi/sheet02_lambda/`: Arduino Uno and the custom chip `lambda-probe`. Pin 9 sends the fuel command as PWM (`analogWrite`, 490 Hz, duty = $q/2$), A0 reads the sensor voltage 02, A1 the ground truth $\text{LAM} = \lambda/2$. The chip's sliders set the unmetered air (slider – 10 in %, default +5 %) and the transport delay (default 200 ms).

- Implement the two-point controller with integrator ($r = 0.2/\text{s}$) in `loop()`. Start the simulation and observe the limit cycle in the Serial Monitor (`u_o2`, `lambda`, `q`). Copy 5 s of data into JSLinux and plot λ and U ; compare with Fig. 1.3 of the script and with your simulation of Exercise 2.3 c).
- Measure the oscillation period (column `period_ms`) for transport delays of 100, 200, 400 and 800 ms and plot period over T_d . Determine slope and intercept and interpret them.
- Move the air slider from +5 % to –10 % and to +10 %. Describe the transient and the new mean value of q .
- Add the jump $\text{PJUMP} = 0.03$, then reduce `RATE` to 0.1 with $\text{PJUMP} = 0.025$. Compare period and λ amplitude. Why can no two-point controller make the oscillation disappear, and what would a continuous (PI) lambda controller need?

Deliverables

2.1: property table, proofs. 2.2: test program, property table with numerical evidence, $h[n]$ of the LTI systems. 2.3: program, values of τ , response times, prediction formula and comparison table. 2.4: sketch, plot of λ and U over 5 s, period over T_d (table + plot), results of c) and d).